MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER

Discipline Specific Elective (DSE) - MG4DSEPHY205 - THEORY OF RELATIVITY

Duration: 2 hours **Maximum Marks: 70**

PART – A

1. Velocity composition laws

$$u_x^{'}=\frac{u_x-v}{1-\frac{u_x v}{c^2}}\, u_y^{'}=\frac{u_y\sqrt{1-\frac{v^2}{c^2}}}{1-\frac{u_x v}{c^2}}, u_z^{'}=\frac{u_z\sqrt{1-\frac{v^2}{c^2}}}{1-\frac{u_x v}{c^2}}$$

Special case:
When $u_x=c$, $u_y=0$, $u_z=0$

$$u_x^{'}=c, u_y^{'}=0, u_z^{'}=0$$

Scheme: 3 marks for correct answer; 2 marks for velocity rules or relevant expressions

2. Particle with uniform velocity $v=1$ (speed of light, $c=1$)
(Space-time diagram representation of motion)

Scheme: 3 marks for correct answer; 2 for relevant space time diagram

3. Velocity of light for observer R
No. If velocity of light is c observer R, then for observer $R^{'}$:

$$u'=c-v$$

Scheme: 3 marks for correct answer; 2 marks for relevant expressions

4. Tensor transformation

For a $(2,0)$ tensor:

$$T^{'\mu\nu}=\Lambda^{\mu}{}_{\rho}\Lambda^{\nu}{}_{\sigma}T^{\rho\sigma}$$

For a $(0,2)$ tensor:

$$S_{\mu\nu}^{'}=\Lambda^{\rho}{}_{\mu}\Lambda^{\sigma}{}_{\nu}S_{\rho\sigma}$$

Outer product:

$$T^{\mu\nu} S_{\kappa\lambda}$$

Scheme: 3 marks for correct answer; 2 marks for relevant expressions

5. Four-vector notation

$$A^{\mu} = (A^0, A^1, A^2, A^3)$$

Lowering index:

$$A_{\mu} = \eta_{\mu\nu} A^{\nu} = (-A^0, A^1, A^2, A^3)$$

Scheme: 3 marks for correct answer; 2 marks for relevant expressions

6. Rotation matrix

$$\begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Inverse rotation:

$$\begin{bmatrix} \cos\theta & \sin\theta & 0 \\ -\sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Rotation about z-axis (anticlockwise direction)

Scheme: 3 marks for correct answer; 2 marks for relevant expressions

7. Spacetime interval

$$ds^2 = -dt^2 + dx^2 + dy^2 + dz^2 \quad (c = 1)$$

Classification:

- $ds^2 < 0 \rightarrow$ timelike
- $ds^2 > 0 \rightarrow$ spacelike
- $ds^2 = 0 \rightarrow$ lightlike

Scheme: 3 marks for correct answer; 2 marks for relevant expressions

8. Unit vectors in polar coordinates

$$\vec{e}_r = \cos\theta\ \vec{e}_x + \sin\theta\ \vec{e}_y$$

Scheme: 3 marks for correct answer; 2 marks for relevant expressions

9. Tidal forces

$$\frac{d^2\xi^\mu}{d\tau^2} = R^\mu{}_{\rho\nu\sigma}\, U^\rho U^\nu\, \xi^\sigma$$

For two geodesics that differ by ξ^μ, the tidal force has the form above with U^μbeing the tangent along the geodesic.

Scheme: 3 marks for correct answer; 2 marks for relevant expressions

10.

$$\vec{e}_r = \vec{e}_r \vec{e}_\theta = \frac{1}{r}\,\vec{e}_\theta$$

Scheme: 3 marks for correct answer; 2 marks for relevant expressions

11.

$$D_\nu V^\mu = \partial_\nu V^\mu + \Gamma^\mu_{\nu\rho} V^\rho$$

Scheme: 3 marks for correct answer; 2 marks for relevant expressions

12.
Units with Newton's gravitational constant:

$$G_N = 1$$

and speed of light:

$$c = 1$$

Scheme: 3 marks for correct answer; 2 marks for relevant expressions

13.

$$-t^2 + x^2 + y^2 + z^2 = -9 + 4 + 0 + 0 = -5 \quad t^2 - x^2 - y^2 - z^2 = 5$$

Scheme: 3 marks for correct answer; 2 marks for relevant expressions

14. (Lorentz transformation)

$$t' = \frac{t - \frac{v}{c^2}x}{\sqrt{1 - \frac{v^2}{c^2}}},\; x' = \frac{x - vt}{\sqrt{1 - \frac{v^2}{c^2}}},\; y' = y,\; z' = z$$

For $v \ll c$:

$$\sqrt{1 - \frac{v^2}{c^2}} \approx 1,\; \frac{v}{c} \approx 0 \Rightarrow t' = t,\; x' = x - vt,\; y' = y,\; z' = z$$

Scheme: 3 marks for correct answer; 2 marks for relevant expressions

15.

$$E^2 = p^2c^2 + m^2c^4$$

For low momentum $(p \ll mc)$:

$$E = mc^2\sqrt{1+\frac{p^2}{m^2c^2}}$$

Using expansion:

$$E \approx mc^2 + \frac{p^2}{2m}$$

Time dilation example:

Rest frame R:
Time for 100 flashes:

$$\Delta\tau = 2 \text{ min} = 120 \text{ s}$$

Moving frame R':

$$\Delta t = \frac{\Delta\tau}{\sqrt{1-\frac{v^2}{c^2}}}$$

For $v = 0.6c$:

$$\Delta t = \frac{1}{\sqrt{1-0.36}} = \frac{1}{0.8} = 1.25$$

Frequency:

$$¿\frac{100}{1.25} = 80 \text{ flashes/min}$$

Scheme: 3 marks for correct answer; 2 marks for relevant expressions

16.

$$E^2 = p^2c^2 + m^2c^4 \quad E = \sqrt{p^2c^2 + m^2c^4}$$

For $p \ll mc$:

$$E \approx m c^2 + \frac{p^2}{2m}$$

Scheme: 3 marks for correct answer; 2 marks for relevant expressions

17.

For a light-like 4-vector k^μ:

$$k_\mu k^\mu = \eta_{\mu\nu} k^\mu k^\nu = -k_0^2 + k_1^2 + k_2^2 + k_3^2 = 0$$

Scheme: 3 marks for correct answer; 2 marks for relevant expressions

18.

A symmetric tensor $S_{\mu\nu}$, where $\mu, \nu = 0,1,2,3$:

Total components = $4 \times 4 = 16$

Since symmetric:

$$S_{\mu\nu} = S_{\nu\mu}$$

Independent components:

- Off-diagonal: $\frac{4 \times 3}{2} = 6$
- Diagonal: 4

$$\text{Total} = 6 + 4 = 10$$

Scheme: 3 marks for correct answer; 2 marks for relevant expressions

19. (Christoffel symbols symmetry)

$$\Gamma^\mu_{\rho\sigma} = \Gamma^\mu_{\sigma\rho}$$

Scheme: 3 marks for correct answer; 2 marks for relevant expressions

20.

$$\frac{\partial \vec{e}_x}{\partial x} = 0, \frac{\partial \vec{e}_y}{\partial y} = 0$$

Scheme: 3 marks for correct answer; 2 marks for relevant expressions

21.

Parallel transport of a vector along a curve computes the rate of change of the vector as one moves from one spacetime point to another while keeping it parallel.

For vector $V = V^\mu \vec{e}_\mu$:

$$D_\nu V^\mu = \partial_\nu V^\mu + \Gamma^\mu_{\nu\rho} V^\rho$$

Scheme: 3 marks for correct answer; 2 marks for relevant expressions

22.
Riemann tensor $R_{\mu\nu\rho\sigma}$:

Total components:

$$4^4 = 256 \left(\text{but with symmetries reduced} \right)$$

Using antisymmetry and symmetry properties:

- Reduced to 20 components
- With additional identities → further reduced

Finally:

$$\text{Independent components} = 5$$

(Bianchi identity used):

$$D_\lambda R_{\mu\nu\rho\sigma} + D_\rho R_{\mu\nu\sigma\lambda} + D_\sigma R_{\mu\nu\lambda\rho} = 0$$

$$R_{\mu\nu\rho\sigma} = R_{\rho\sigma\mu\nu}$$

Thus:

$$R_{¿\,¿}$$

After all symmetry + Bianchi constraints:

$$5 \text{ independent components}$$

Scheme: 3 marks for correct answer; 2 marks for relevant expressions